46110
Fundamentals of Aerodynamics

Lecture 7:

- **Downwash and induced drag**
- **Prandtl's lifting-line equation**
- **The elliptic wing**
- **Lifting line technique for general circulation distribution**
- **Start on assignment 2**

DTU Wind Energy
Department of Wind Energy

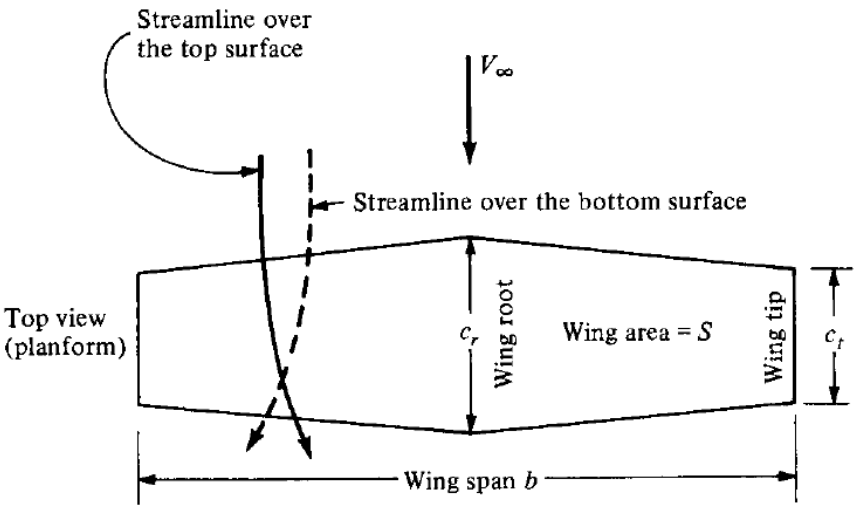
The basic idea in the Prandtl-school of aerodyncamics

The aerodynamic consideration of a wing can be split in two parts:

1. Study of wing sections (2D)
2. Modification of local 2D behavior to account for the complete finite wing

Up to now you have been doing #1, so now we're digging into #2.

So how does the 3D case differ from the 2D case?



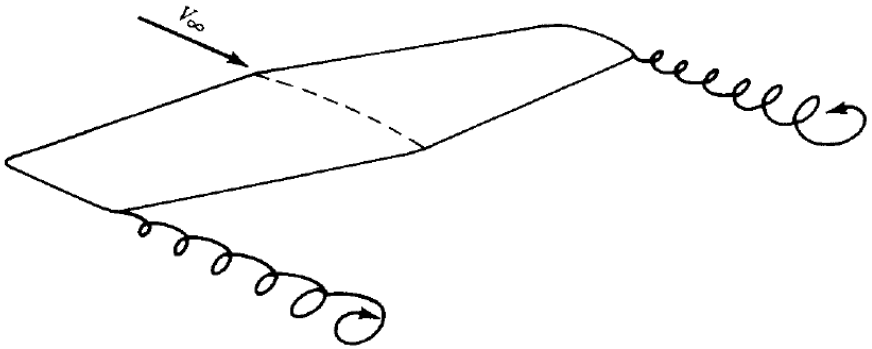


Figure 5.2 Schematic of wing-tip vortices.

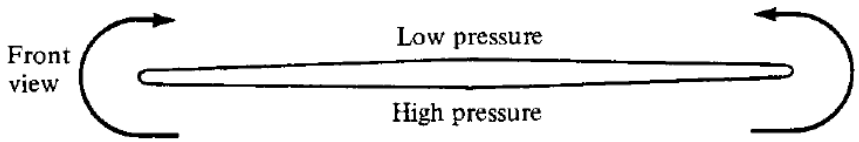


Figure 5.1 Finite wing. In this figure, the curvature of the streamlines over the top and bottom of the wing is exaggerated for clarity.

First-order effect:

Trailing vorticity

↓

Downwash!

The (local) effect of the downwash

The effect of the induced downwash is a reduction of the percieved effective angle of attack:

$$\alpha_{eff} = \alpha - \alpha_i$$

where $\alpha_i = \tan^{-1}\frac{w}{V_\infty} \approx \frac{w}{V_\infty}$

Since the effective lift force acts in a direction perpendicular to the local relative wind, there will be a component of this in the direction of the "global" incoming wing, ie a drag due to lift; induced drag:

$$D'_i = L' \sin\alpha_i \approx L'\alpha_i$$

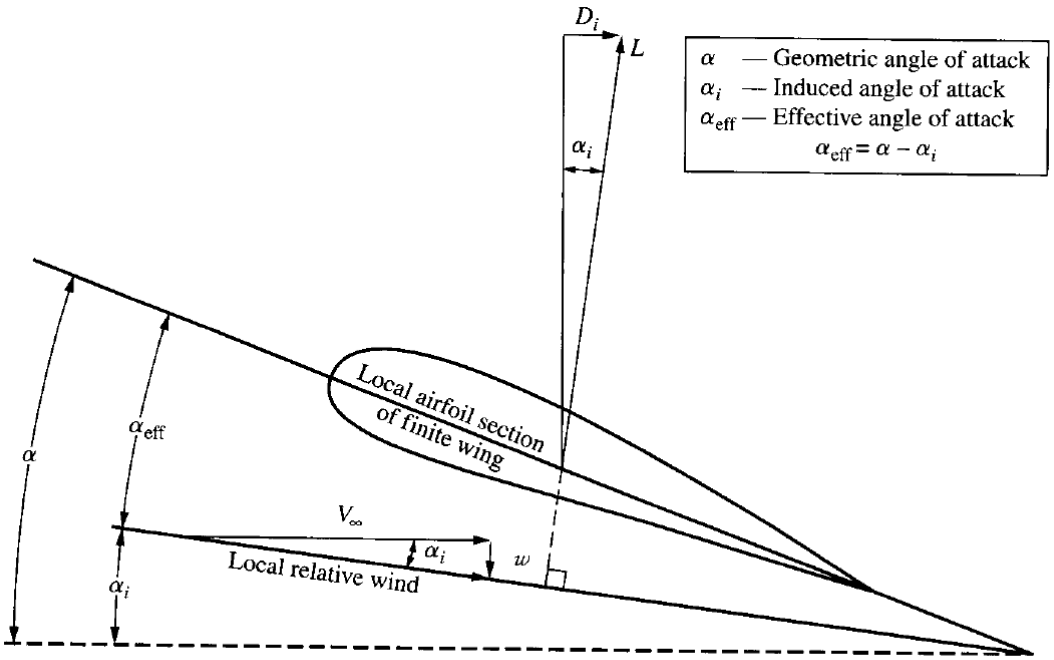
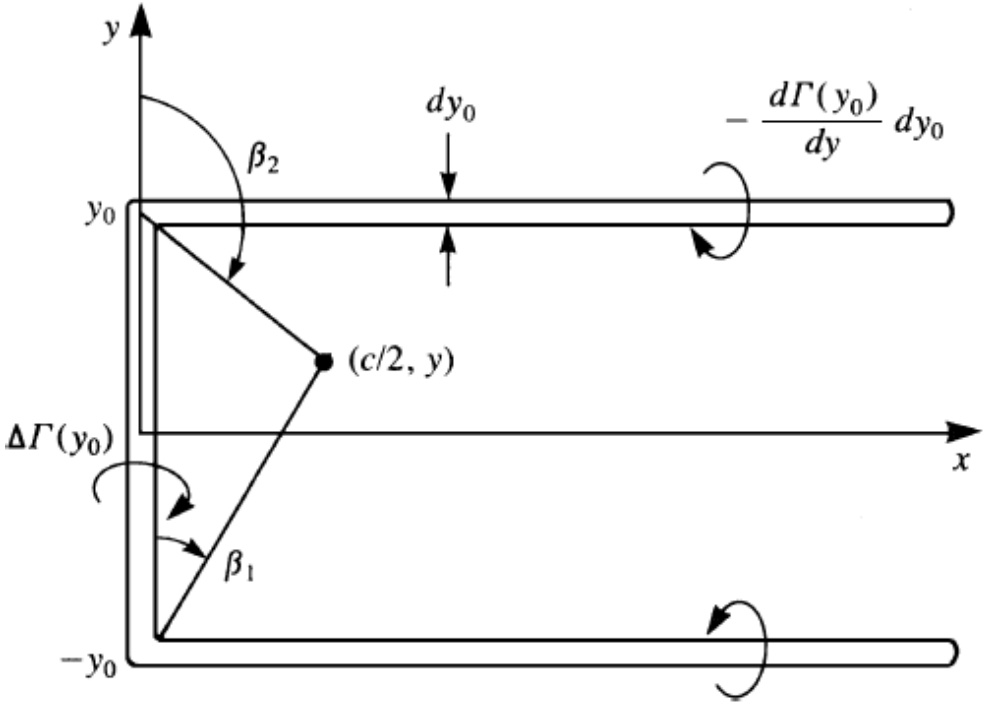


Figure 5.4 Effect of downwash on the local flow over a local airfoil section of a finite wing.

Note that in real life the total drag for an airfoil section include also the contribution from viscous effects (pressure and friction drag=profile drag), which can be obtained from 2D airfoil characteristics.

The horseshoe vortex model

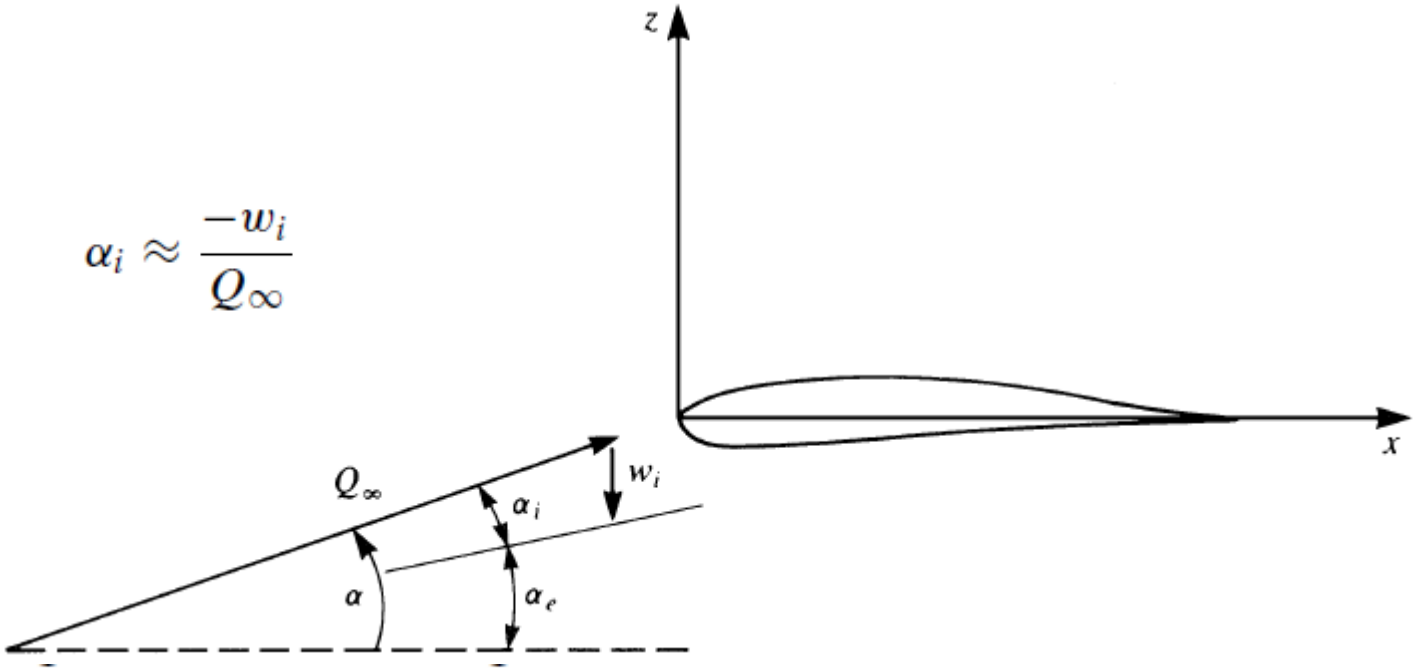


Velocity induced by the segments of a typical horseshoe element

$$w(y) = \frac{\Delta\Gamma(y_0)}{4\pi}\frac{1}{(y-y_0)} \qquad w_i = \frac{1}{4\pi}\int_{-b/2}^{b/2}\frac{[-d\Gamma(y_0)/dy]\,dy_0}{y-y_0}$$

Induced angle of attack

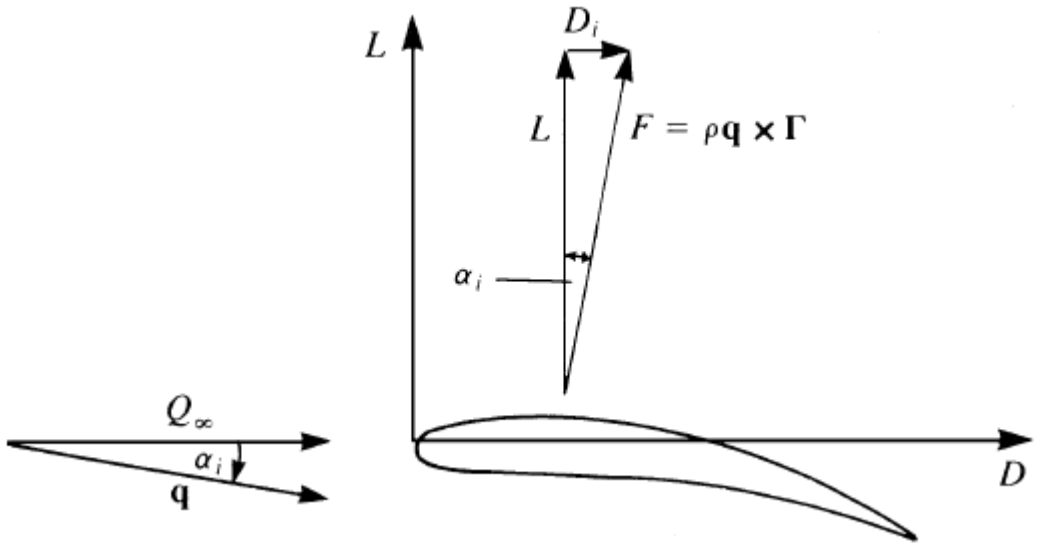
$$\alpha_i \approx \frac{-w_i}{Q_\infty}$$



$$w_i = \frac{1}{4\pi} \int_{-b/2}^{b/2} \frac{[-d\Gamma(y_0)/dy]\, dy_0}{y - y_0}$$

$$\alpha_i = \frac{1}{4\pi Q_\infty} \int_{-b/2}^{b/2} \frac{[d\Gamma(y_0)/dy]\, dy_0}{y - y_0}$$

Induced drag



Assume:
$\cos\alpha_i \approx 1$ and $\sin\alpha_i \approx \alpha_i$

$$L = \rho Q_\infty \int_{-b/2}^{b/2} \Gamma(y)\,dy$$

$$D_i = \rho Q_\infty \int_{-b/2}^{b/2} \alpha_i(y)\Gamma(y)\,dy$$

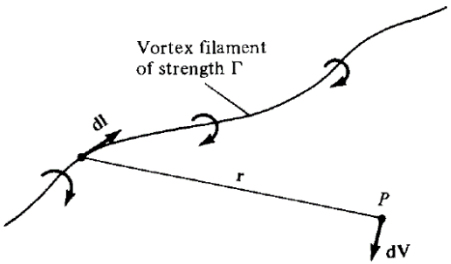
$$\alpha_i \approx \frac{-w_i}{Q_\infty}$$

$$D_i = -\rho \int_{-b/2}^{b/2} w_i(y)\Gamma(y)\,dy$$

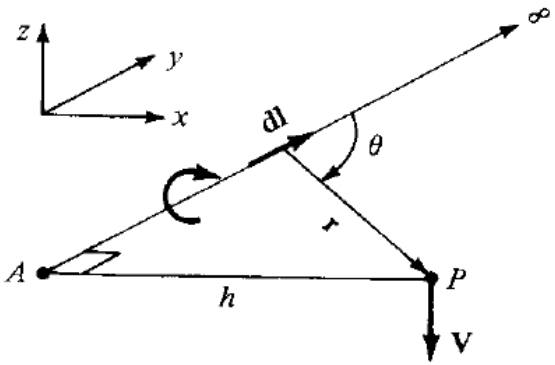
Needed aerodynamic tools

General induced velocity from a vortex filament (Biot-Savart law)

$$\mathbf{dV} = \frac{\Gamma}{4\pi} \frac{\mathbf{dl} \times \mathbf{r}}{|\mathbf{r}|^3}$$



For a semi-infinite straight filament we get: (RH rule for direction)



$$V = \frac{\Gamma}{4\pi h}$$

Needed aerodynamic tools

Helmholz's vortex theorems:

1) The strength of a vortex filament is constant along its length
2) A vortex filament cannot end in a fluid; it must extend to the boundaries of the fluid (possibly to infinity) or form a closed path.

Kutta-Joukowski theorem:

The relation between circulation, density, velocity and force per spanlength

$$\bar{L}'_{rel} = \rho \bar{V}_{rel} \times \bar{\Gamma}$$

which for the lifting part (per spanlength) of a nonyawed wing is

$$L' = \rho V_\infty \Gamma$$

And the induced drag part (per spanlength) of a nonyawed wing is

$$D' = \rho w \Gamma$$

ASSUMPTION OF SMALL VARIATION OF DOWNWASH OVER THE CHORD: SLENDER WING ASSUMED

What we want

Is to calculate:

- ✓**Local induced drag**
- ✓**Local lift**
- ✓**Total induced drag**
- ✓**Total lift**

So in the process we'll need to determine

- ✓**Downwash**
- ✓**Induced angle of attack**
- ✓**Circulation distribution**

Computation of downwash

The "vortex system": superposition of horseshoe vortices

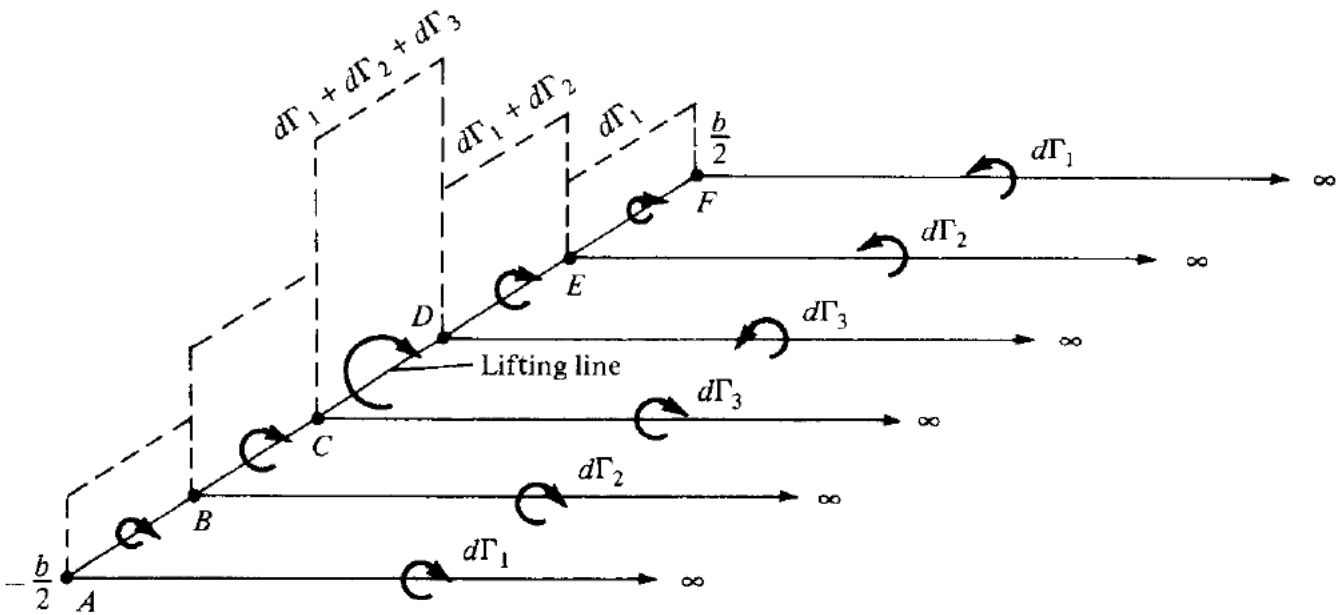


Figure 5.12 Superposition of a finite number of horseshoe vortices along the lifting line.

Computation of downwash

The "vortex system": superposition of horseshoe vortices…infinitely many

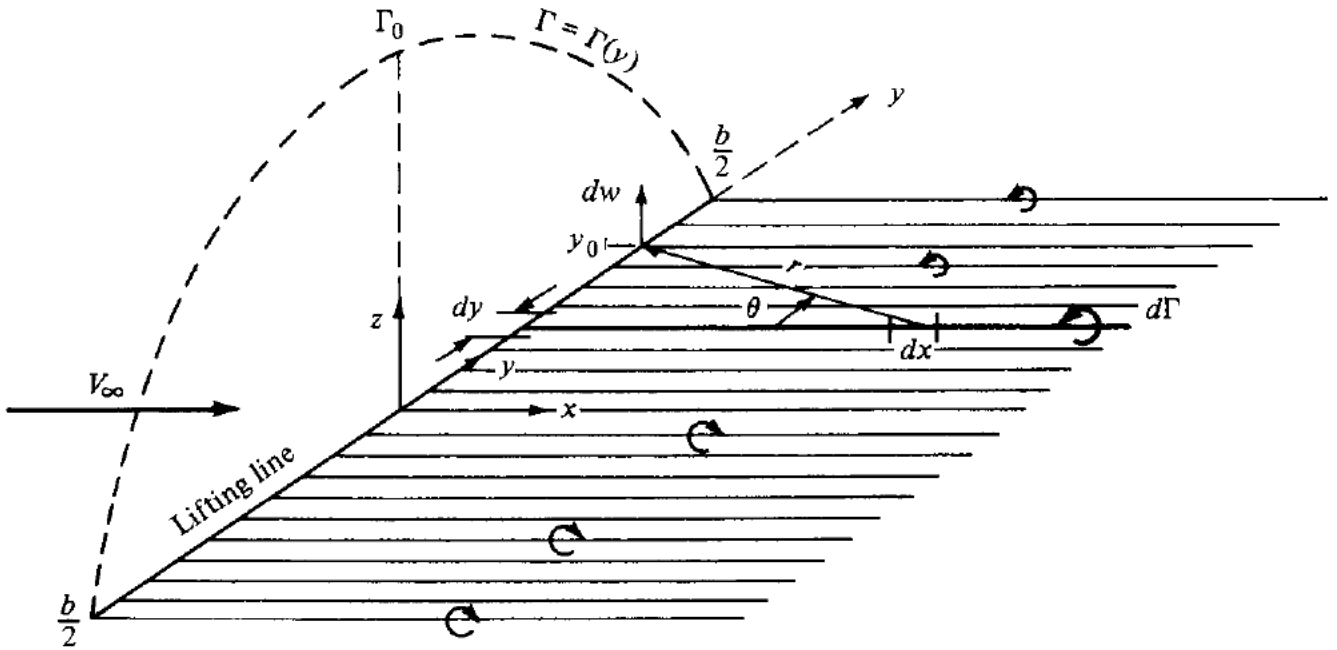


Figure 5.13 Superposition of an infinite number of horseshoe vortices along the lifting line.

Computation of downwash

From the semiinfinite vortex filament induction earlier we get

$$dw = -\frac{(d\Gamma/dy)\,dy}{4\pi(y_0 - y)}$$

(OBS: despite the name, w is positive upward)

So the total downwash at y_0 can be obtained from integration

$$\boxed{w(y_0) = -\frac{1}{4\pi}\int_{-b/2}^{b/2}\frac{(d\Gamma/dy)\,dy}{y_0 - y}}$$

With small angles and w positive up we get

$$\boxed{\alpha_i(y_0) = \frac{1}{4\pi V_\infty}\int_{-b/2}^{b/2}\frac{(d\Gamma/dy)\,dy}{y_0 - y}}$$

Prandtl's fundamental equation for LL theory

If (unstalled) "2D" lift coeff is

$$c_l = a_0[\alpha_{\text{eff}}(y_0) - \alpha_{L=0}] = 2\pi[\alpha_{\text{eff}}(y_0) - \alpha_{L=0}]$$

Then we can equate the KJ and the 2D lift to obtain

$$c_l = \frac{2\Gamma(y_0)}{V_\infty c(y_0)}$$

Which combined with the top equation results in

$$\alpha_{\text{eff}} = \frac{\Gamma(y_0)}{\pi V_\infty c(y_0)} + \alpha_{L=0}$$

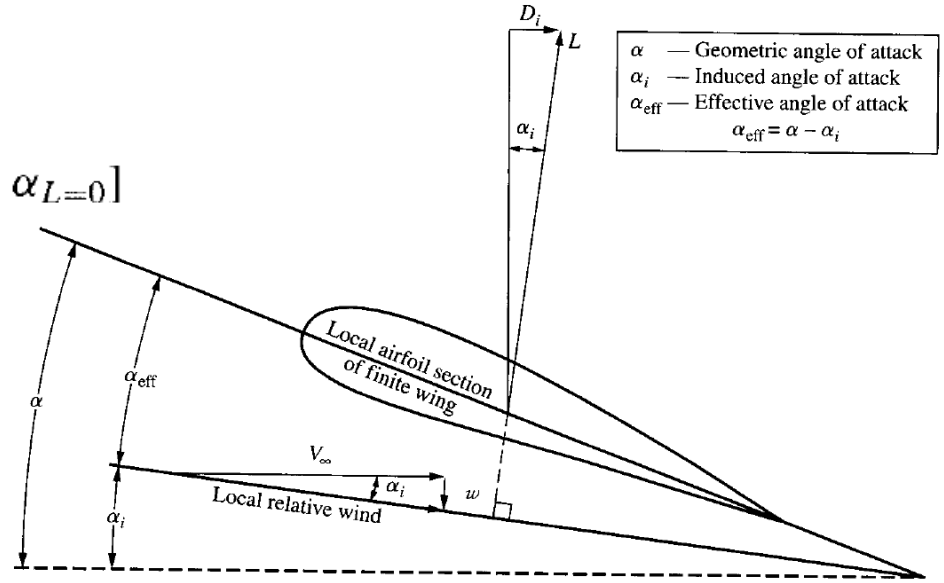


Figure 5.4 Effect of downwash on the local flow over a local airfoil section of a finite wing.

Now remember from previously, that $\alpha_{\text{eff}} = \alpha - \alpha_i$

Prandtl's fundamental equation for LL theory

Combining the last two equations for the effective angle of attack with 5.18 (calculation of the induced angle of attack) we finally arrive at the ***fundamental equation of Prandtl's lifting-line theory:***

$$\boxed{\alpha(y_0) = \frac{\Gamma(y_0)}{\pi V_\infty c(y_0)} + \alpha_{L=0}(y_0) + \frac{1}{4\pi V_\infty} \int_{-b/2}^{b/2} \frac{(d\Gamma/dy)\,dy}{y_0 - y}}$$

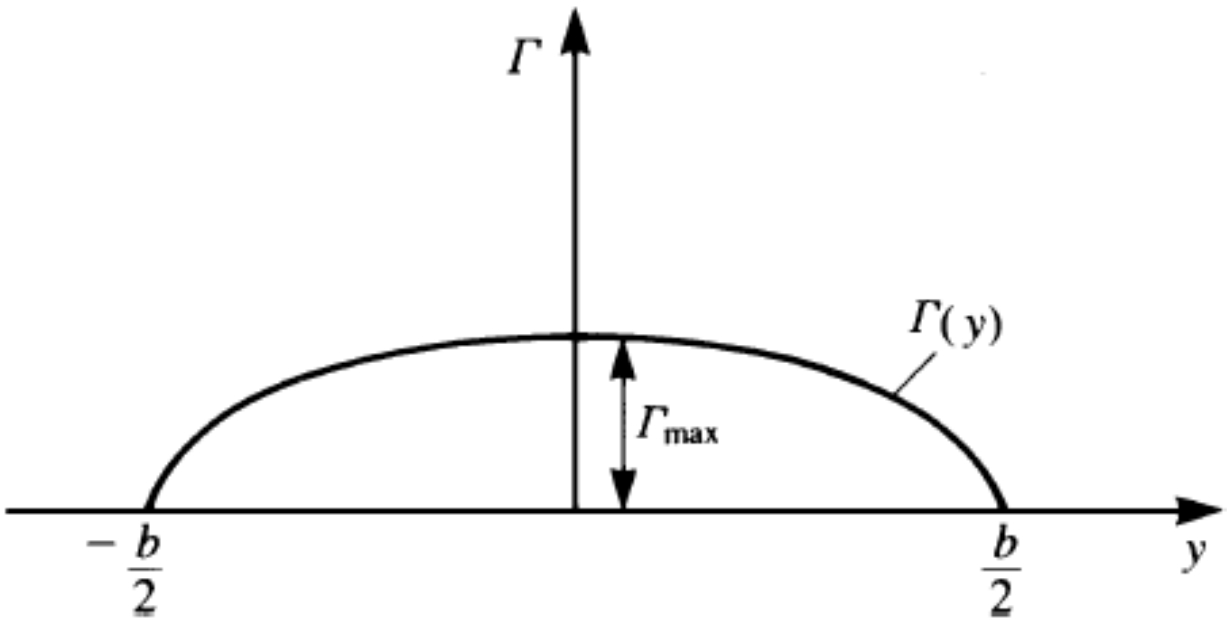
In words:

"The geometric angle of attack needs to be equal to the sum of the effective angle plus the induced angle of attack".

It is an integro-differential equation in the unknown circulation.

The Elliptic lift distribution

$$\Gamma(y) = \Gamma_{\max}\left[1 - \left(\frac{y}{b/2}\right)^2\right]^{1/2}$$



The Elliptic lift distribution

$$\Gamma(y) = \Gamma_{\max}\left[1 - \left(\frac{y}{b/2}\right)^2\right]^{1/2}$$

$$\frac{d\Gamma(y)}{dy} = \frac{\Gamma_{\max}}{2}\left[1 - \left(\frac{y}{b/2}\right)^2\right]^{-1/2}\left(-2\frac{4}{b^2}y\right)$$

$$w_i = \frac{1}{4\pi}\int_{-b/2}^{b/2} \frac{[-d\Gamma(y_0)/dy]\,dy_0}{y - y_0}$$

$$w_i(y) = \frac{\Gamma_{\max}}{\pi b^2}\int_{-b/2}^{b/2}\left[1 - \left(\frac{y_0}{b/2}\right)^2\right]^{-1/2}\frac{y_0}{y - y_0}\,dy_0$$

The Elliptic lift distribution

transformation

$$y = \frac{b}{2}\cos\theta \qquad \Gamma(y) = \Gamma_{\max}\left[1 - \left(\frac{y}{b/2}\right)^2\right]^{1/2}$$

$$dy = -\frac{b}{2}\sin\theta\, d\theta \qquad \Gamma(\theta) = \Gamma_{\max}[1 - \cos^2\theta]^{1/2} = \Gamma_{\max}\sin\theta$$

$$w_i(y) = \frac{\Gamma_{\max}}{\pi b^2}\int_{-b/2}^{b/2}\left[1 - \left(\frac{y_0}{b/2}\right)^2\right]^{-1/2}\frac{y_0}{y - y_0}\,dy_0$$

$$w_i = \frac{\Gamma_{\max}}{\pi b^2}\int_{\pi}^{0}[1 - \cos^2\theta_0]^{-1/2}\frac{(b/2)\cos\theta_0[(-b/2)\sin\theta_0]d\theta_0}{(b/2)(\cos\theta - \cos\theta_0)}$$

Recall: $\displaystyle\int_0^{\pi}\frac{\cos n\theta_0}{\cos\theta_0 - \cos\theta}\,d\theta_0 = \frac{\pi\sin n\theta}{\sin\theta}, \quad n = 0, 1, 2, \ldots$

$$w_i = \frac{-\Gamma_{\max}}{2\pi b}\int_0^{\pi}\frac{\cos\theta_0\, d\theta_0}{(\cos\theta_0 - \cos\theta)} = \frac{-\Gamma_{\max}}{2\pi b}\frac{\pi\sin\theta}{\sin\theta}$$

$$\boxed{\begin{aligned} w_i &= -\frac{\Gamma_{\max}}{2b} \\ \alpha_i &= \frac{\Gamma_{\max}}{2bQ_\infty} \end{aligned}}$$

The Elliptic lift distribution

$$\int_{-b/2}^{b/2} \Gamma(y)\,dy = \frac{\pi}{2}\Gamma_{\max}\frac{b}{2} = \frac{\pi b}{4}\Gamma_{\max} \qquad \text{(half ellipse area)}$$

Consequently, the lift and the drag of the wing can be evaluated as

$$L = \rho Q_\infty \int_{-b/2}^{b/2} \Gamma(y)\,dy = \frac{\pi b}{4}\rho Q_\infty \Gamma_{\max}$$

$$D_i = \rho Q_\infty \int_{-b/2}^{b/2} \alpha_i \Gamma(y)\,dy = \alpha_i L = \left(\frac{\Gamma_{\max}}{2bQ_\infty}\right)\frac{\pi b}{4}\rho Q_\infty \Gamma_{\max} = \frac{\pi}{8}\rho \Gamma_{\max}^2$$

$$\boxed{\begin{aligned} C_L &\equiv \frac{L}{(1/2)\rho Q_\infty^2 S} = \frac{\pi}{2}\frac{b}{S}\frac{\Gamma_{\max}}{Q_\infty} \\ C_{D_i} &\equiv \frac{D_i}{(1/2)\rho Q_\infty^2 S} = \frac{\pi}{4S}\frac{\Gamma_{\max}^2}{Q_\infty^2} = \frac{1}{\pi}\frac{S}{b^2}C_L^2 \end{aligned}}$$

The Elliptic wing

local lift slope of m_0

local effective angle of attack is α_e

$$\Rightarrow C_l(y) = \frac{\rho Q_\infty \Gamma(y)}{(1/2)\rho Q_\infty^2 c(y)} = m_0(y)\alpha_e(y)$$

Recall Prandtl's lifting-line equation:

$$\alpha_e = \alpha - \alpha_i - \alpha_{L0} \Rightarrow \frac{-2\Gamma(y)}{m_0(y)c(y)Q_\infty} - \frac{1}{4\pi Q_\infty}\int_{-b/2}^{b/2} \frac{[d\Gamma(y_0)/dy]\,dy_0}{y - y_0} + \alpha(y) - \alpha_{L0}(y) = 0$$

$$\Rightarrow \quad \frac{-2\Gamma_{\max}}{m_0(y)c(y)Q_\infty}\left[1 - \left(\frac{y}{b/2}\right)^2\right]^{1/2} - \frac{\Gamma_{\max}}{2bQ_\infty} + \alpha(y) - \alpha_{L0}(y) = 0$$

Assume elliptic plan form: $c(y) = c_0\left[1 - \left(\frac{y}{b/2}\right)^2\right]^{1/2}$

$$\Rightarrow \quad \frac{-2\Gamma_{\max}}{m_0(y)c_0 Q_\infty} - \frac{\Gamma_{\max}}{2bQ_\infty} + \alpha(y) - \alpha_{L0}(y) = 0$$

constants

$$\Rightarrow \alpha = \alpha(y) = const$$

The Elliptic wing

$$\Gamma_{\max} = \frac{2bQ_\infty(\alpha - \alpha_{L0})}{1 + 4b/m_0 c_0}$$

Area: $S = \pi \frac{c_0}{2}\frac{b}{2}$

Aspect ratio: $\text{Æ} \equiv \frac{b^2}{S}$

$$\Gamma_{\max} = \frac{2bQ_\infty(\alpha - \alpha_{L0})}{1 + \pi \text{Æ}/m_0}$$

From previous:

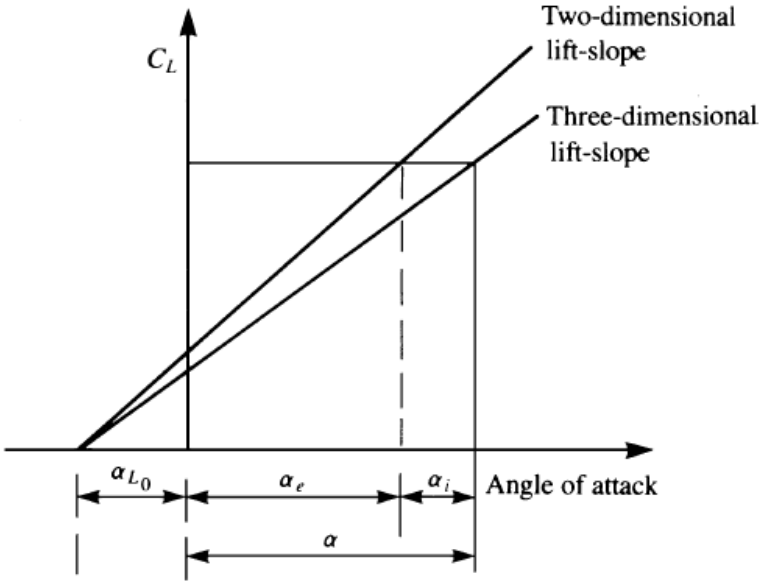
$$C_L \equiv \frac{L}{(1/2)\rho Q_\infty^2 S} = \frac{\pi}{2}\frac{b}{S}\frac{\Gamma_{\max}}{Q_\infty}$$

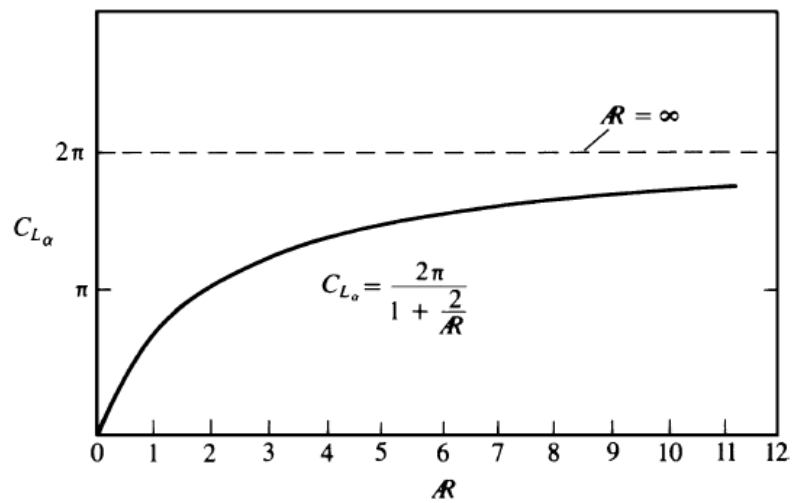
$$C_{D_i} \equiv \frac{D_i}{(1/2)\rho Q_\infty^2 S} = \frac{\pi}{4S}\frac{\Gamma_{\max}^2}{Q_\infty^2} = \frac{1}{\pi}\frac{S}{b^2}C_L^2$$

$$\Rightarrow \boxed{\begin{aligned} C_L &= \frac{2\pi}{1 + 2/\text{Æ}}(\alpha - \alpha_{L0}) \equiv C_{L_\alpha}(\alpha - \alpha_{L0}) \\ C_{D_i} &= \frac{1}{\pi \text{Æ}} C_L^2 \end{aligned}}$$

Where we assumed: $m_0 = 2\pi$

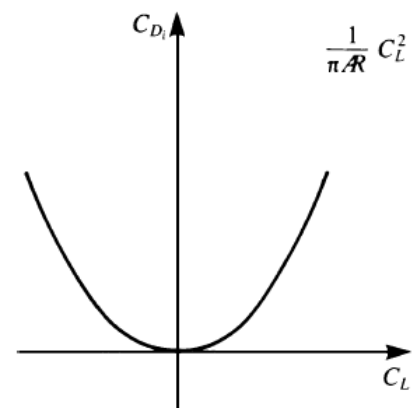
The Elliptic wing



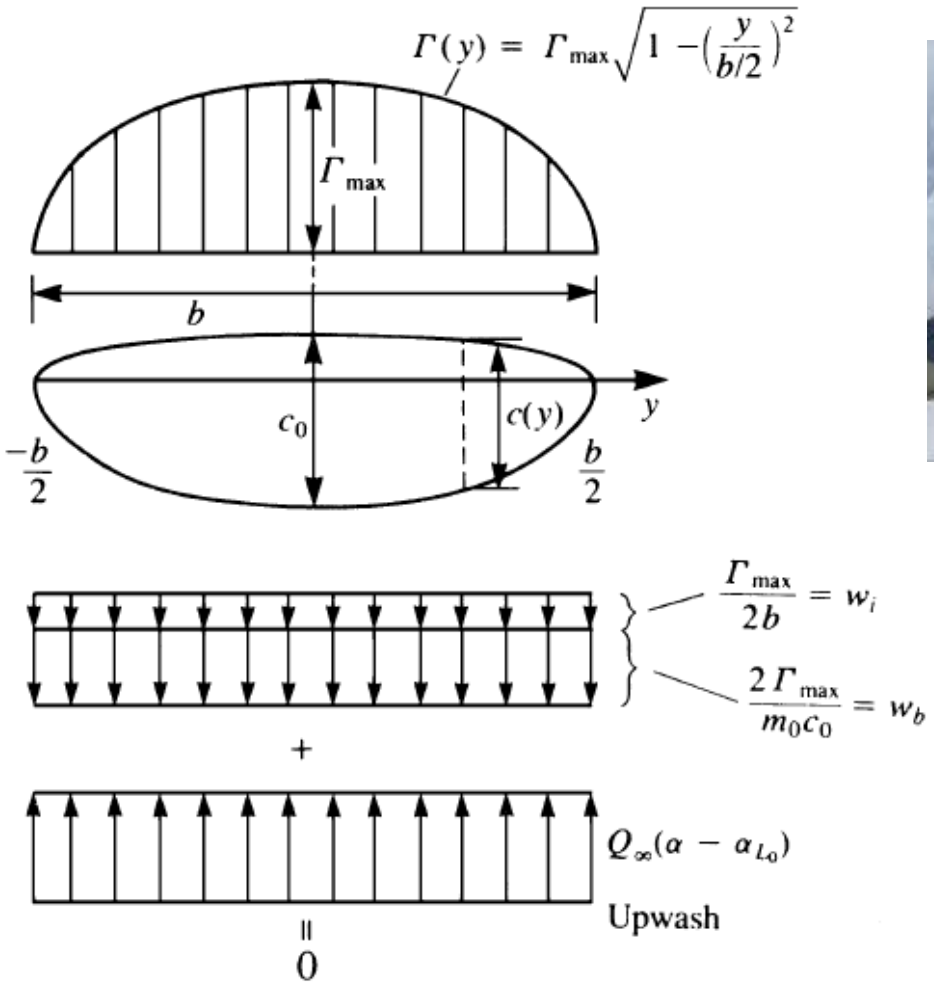


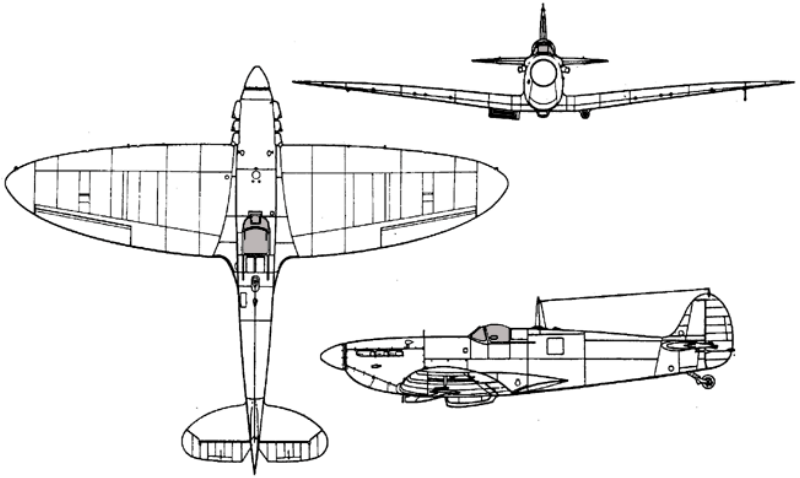
$$C_L = \frac{2\pi}{1 + 2/AR}(\alpha - \alpha_{L0}) \equiv C_{L_\alpha}(\alpha - \alpha_{L0})$$

$$C_{D_i} = \frac{1}{\pi AR} C_L^2$$



The Elliptic wing

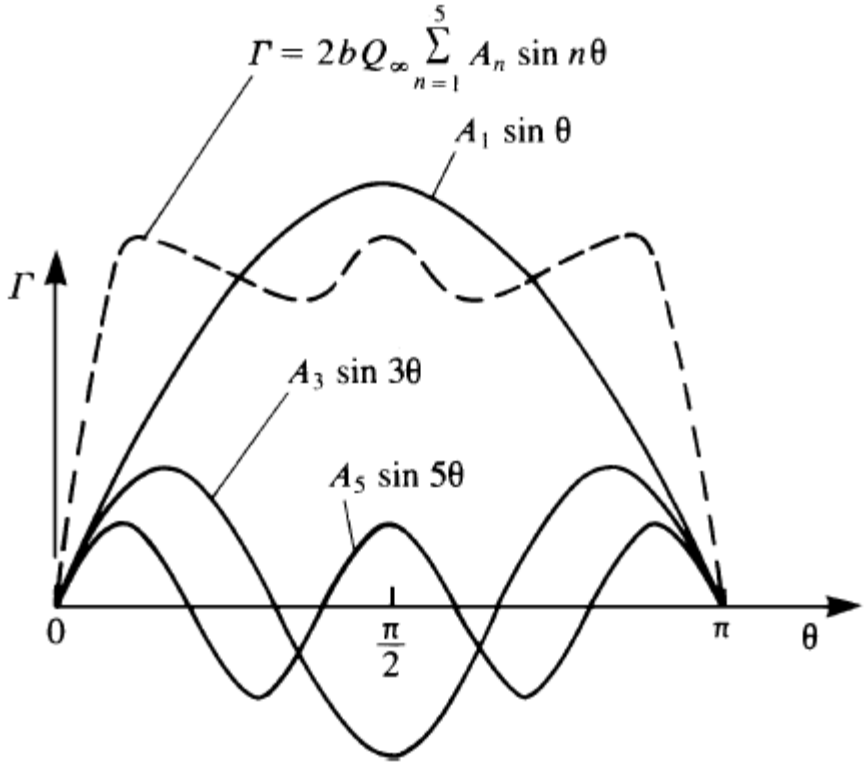




General spanwise circulation distribution

Fourier expansion: $\Gamma(\theta) = 2bQ_\infty \sum_{n=1}^{\infty} A_n \sin n\theta$

Note:
$\Gamma(0) = \Gamma(\pi) = 0$



General spanwise circulation distribution

$$\Gamma(\theta) = 2bQ_\infty \sum_{n=1}^{\infty} A_n \sin n\theta$$

transformation

$$y = \frac{b}{2}\cos\theta$$

$$dy = -\frac{b}{2}\sin\theta\, d\theta$$

Prandtl's lifting-line equation:

$$\frac{-2\Gamma(y)}{m_0(y)c(y)Q_\infty} - \frac{1}{4\pi Q_\infty}\int_{-b/2}^{b/2} \frac{[d\Gamma(y_0)/dy]\,dy_0}{y - y_0} + \alpha(y) - \alpha_{L0}(y) = 0 \quad \Rightarrow$$

$$\frac{-4b}{m_0(\theta)c(\theta)} \sum_{n=1}^{\infty} A_n \sin n\theta$$

$$+ \frac{-b}{2\pi}\int_{\pi}^{0} \frac{\sum_{n=1}^{\infty} A_n n \cos n\theta_0 [1/(-b/2)\sin\theta_0][(-b/2)\sin\theta_0 d\theta_0]}{(b/2)(\cos\theta - \cos\theta_0)}$$

$$+\alpha(\theta) - \alpha_{L0}(\theta) = \frac{-4b}{m_0(\theta)c(\theta)} \sum_{n=1}^{\infty} A_n \sin n\theta - \frac{1}{\pi}\int_0^{\pi} \frac{\sum_{n=1}^{\infty} n A_n \cos n\theta_0\, d\theta_0}{\cos\theta_0 - \cos\theta}$$

$$+\alpha(\theta) - \alpha_{L0}(\theta) = 0$$

General spanwise circulation distribution

Again, exploiting: $$\int_0^{\pi} \frac{\cos n\theta_0}{\cos\theta_0 - \cos\theta}\, d\theta_0 = \frac{\pi \sin n\theta}{\sin\theta}, \quad n = 0, 1, 2, \ldots$$

We get $$\boxed{\frac{-4b}{m_0(\theta)c(\theta)} \sum_{n=1}^{\infty} A_n \sin n\theta - \sum_{n=1}^{\infty} n A_n \frac{\sin n\theta}{\sin\theta} + \alpha(\theta) - \alpha_{L0}(\theta) = 0}$$

Local properties: $$\alpha_i(\theta) = \sum_{n=1}^{\infty} n A_n \frac{\sin n\theta}{\sin\theta}$$

$$C_l = \frac{\rho Q_\infty \Gamma(\theta)}{(1/2)\rho Q_\infty^2 c(\theta)} = \frac{4b}{c(\theta)} \sum_{n=1}^{\infty} A_n \sin n\theta$$

$$C_{d_i} = C_l \alpha_i = \frac{4b}{c(\theta)} \sum_{n=1}^{\infty} A_n \sin n\theta \left(\sum_{k=1}^{\infty} k A_k \frac{\sin k\theta}{\sin\theta} \right)$$

General spanwise circulation distribution

$$C_L = \int_{-b/2}^{b/2} \frac{C_l(y)c(y)\,dy}{S} = \frac{4b}{S}\int_0^{\pi} \sum_{n=1}^{\infty} A_n \sin n\theta \frac{b}{2} \sin\theta\, d\theta$$

$$C_{D_i} = \int_{-b/2}^{b/2} \frac{C_{d_i}(y)c(y)\,dy}{S} = \frac{2b^2}{S}\int_0^{\pi} \sum_{n=1}^{\infty}\sum_{k=1}^{\infty} k A_k A_n \sin k\theta \sin n\theta\, d\theta$$

Recall that

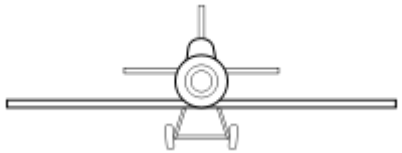
$$\int_0^{\pi} \sin n\theta \sin k\theta\, d\theta = \begin{pmatrix} 0 & \text{for} & n \neq k \\ \pi/2 & \text{for} & n = k \end{pmatrix}$$

$$C_L = \frac{\pi b^2 A_1}{S} = \pi \text{AR} A_1$$

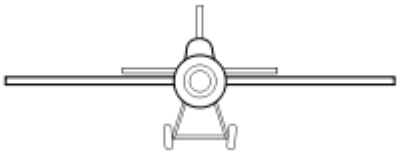
$$C_{D_i} = \frac{\pi b^2}{S}\sum_{n=1}^{\infty} n A_n^2 = \pi \text{AR} \sum_{n=1}^{\infty} n A_n^2 = \frac{C_L^2}{\pi \text{AR}}(1+\delta)$$

where $\delta = \sum_2^N n(A_n/A_1)^2$

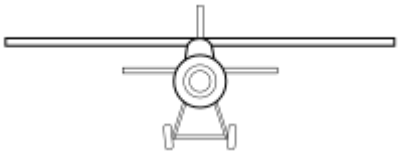
Wing plan forms



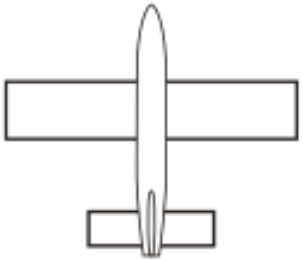
Low wing



Mid wing

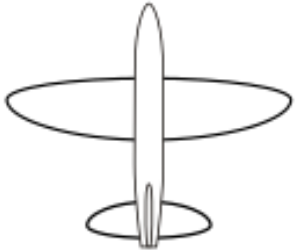


High wing



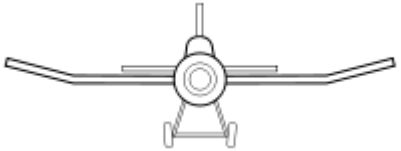
Constant chord

Tapered



Elliptic

Anhedral



Cranked

Dihedral

… and many more configurations (see Wikipedia: ‘Wing Configurations’)

The Spitfire

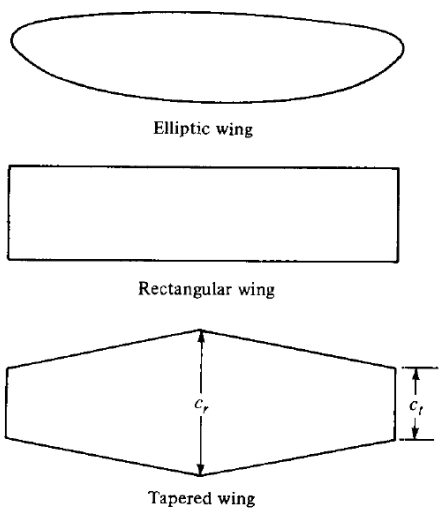
The Spitfire wing may be classified as: “A conventional low-wing cantilever monoplane with unswept elliptical wings of moderate aspect ratio and slight dihedral".

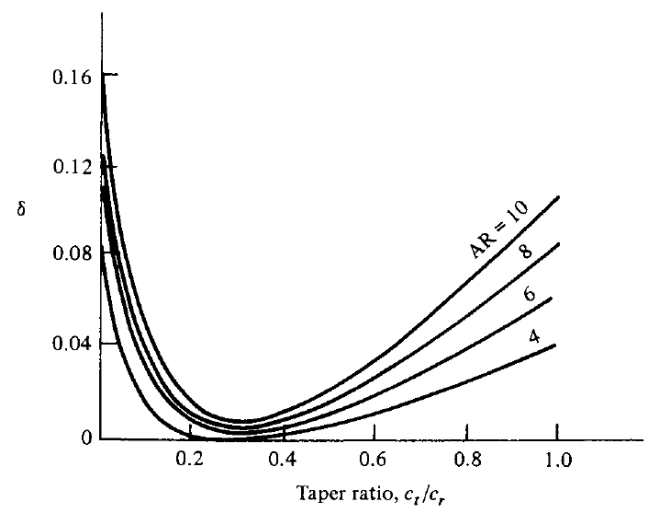
General spanwise circulation distribution

With the definition of the span efficiency factor $e = (1+\delta)^{-1}$ this can finally be written as

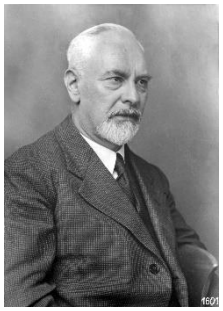
$$\boxed{C_{D,i} = \frac{C_L^2}{\pi e \mathrm{AR}}}$$

Note that $e \leq 1$. Also note that $\delta = 0$ and $e = 1$ for the elliptical lift distribution case; the minimum induced drag case





Ludwig Prandtl:
The father of modern Fluid Mechanics

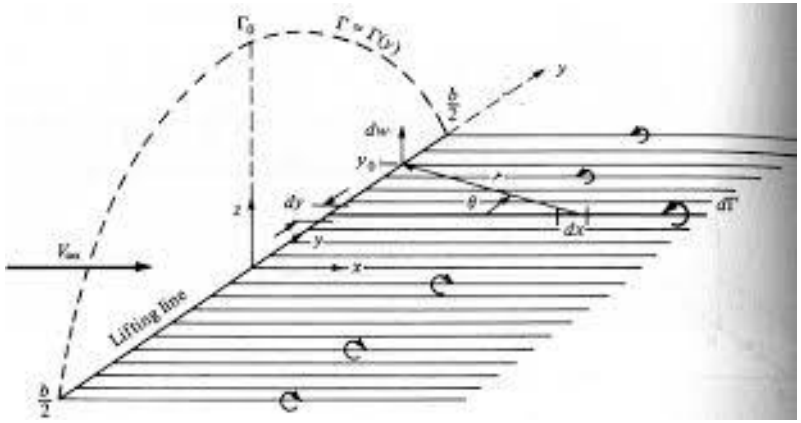


(1875-1953)

Ludwig Prandtl worked at Göttingen until his death on August 15, 1953. His work and achievements in fluid dynamics resulted in equations that simplified understanding, and many are still used today. Therefore many referred to him as the father of modern fluid mechanics.

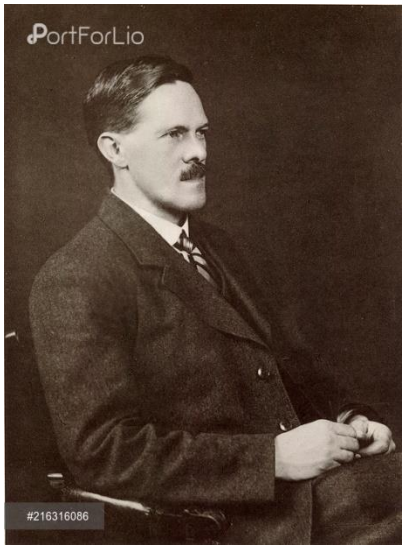
Theories developed and named after Prandtl:

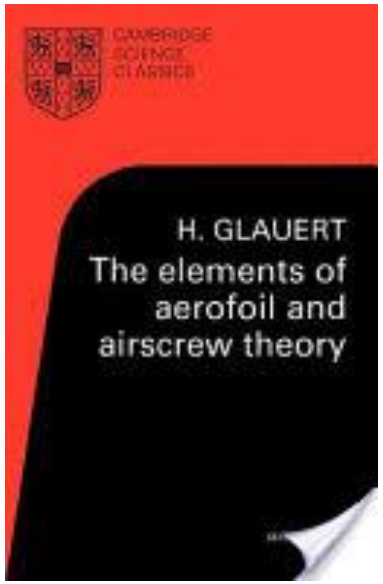
- Prandtl number (heat transfer problems)
- Prandtl-Glauert compressibility correction
- Prandtl's boundary layer equation
- Prandtl's lifting line theory
- Prandtl's law of friction for smooth pipes
- Prandtl-Meyer expansion fans (supersonic flow)
- Prandtl's Mixing Length Concept (theory of turbulence)



Lifting line theory

Hermann Glauert



Hermann Glauert (1892-1934)

- Hermann Glauert was a British aerodynamicist and Principal Scientific Officer of the Royal Aircraft Establishment, Farnborough until his death in 1934.
- Glauert wrote numerous reports and memoranda dealing with aerofoil and propeller theory. His book, The Elements of Aerofoil and Airscrew Theory was the single most important instrument for spreading airfoil and wing theory around the English speaking world.
- Glauert independently developed Prandtl-Glauert method from the then-existing aerodynamic theory and published his results in The Proceedings of the Royal Society in 1928.
- Glauert is the founder of the Blade-Element/Momentum (BEM) theory, which today is widely used for design of wind turbines.